# Molecular Determinants of the binding mechanism between the LTSV40 and the Retinoblastoma protein

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## Background

The short linear motif (SLiM) LxCxE facilitates protein-protein interactions, particularly with the human Retinoblastoma protein (pRb). This motif is found in proteins involved in chromatin regulation, histone modification, and cell cycle control. Among these proteins is the large T antigen of simian virus 40 (LTSV40), which possesses an LxCxE motif at the beginning of an intrinsically disordered region (IDR). This results in its flanking regions being conformationally asymmetric (structured towards its N-terminal region and unstructured towards the C-terminal region).

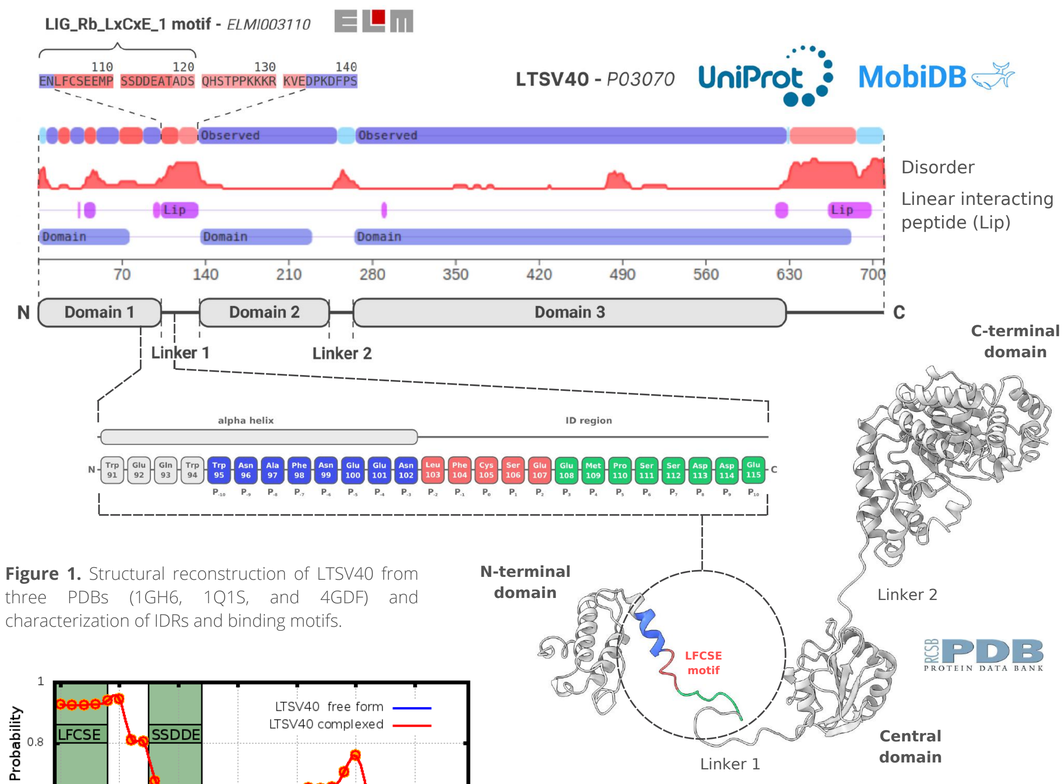


**Figure 1.** Structural reconstruction of LTSV40 from three PDBs (1GH6, 1Q1S, and 4GDF) and characterization of IDRs and binding motifs.

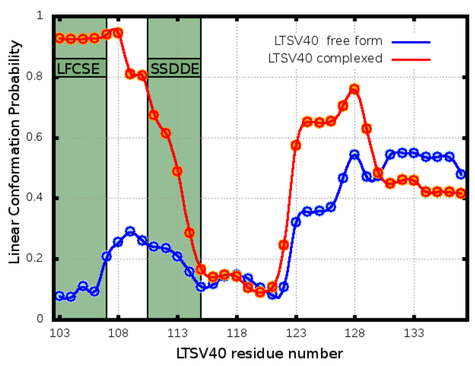


**Figure 6.** Probability of the residues of Linker 1 being within a linear sequence of more than five residues in length. The green rectangles show the LXCXE motif as a conserved pentanucleotide sequence C-terminal to this.

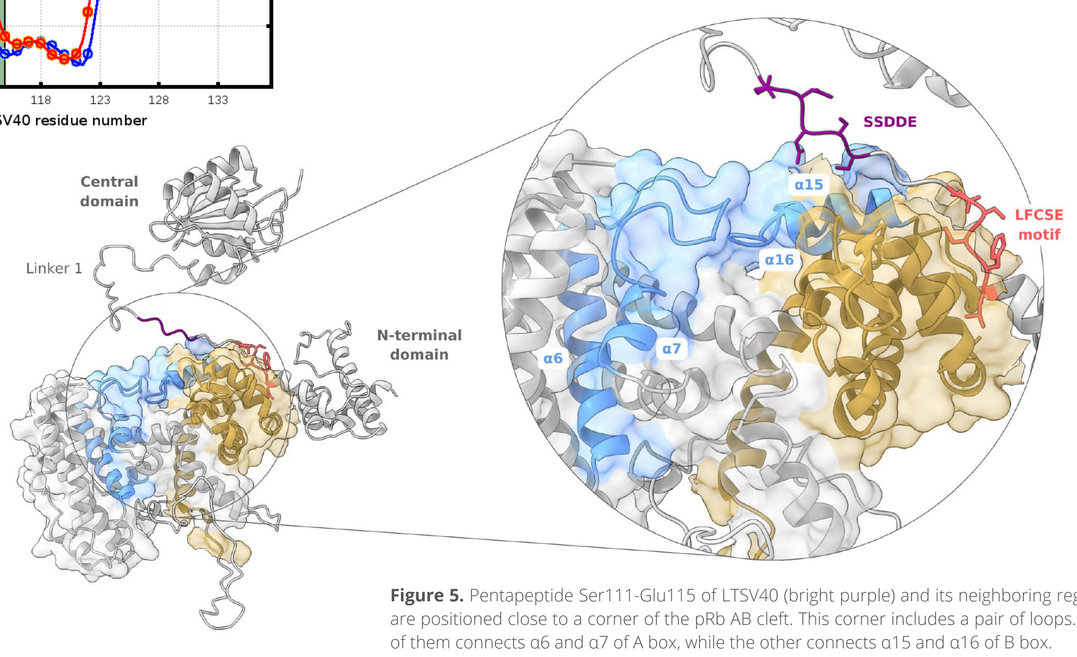


**Figure 4.** LTSV40-pRb complex. Structure of the LxCxE short linear motif (red) from LTSV40 bound to the B-box pocket binding cleft of human pRb (yellow).

**Figure 5.** Pentapeptide Ser111-Glu115 of LTSV40 (bright purple) and its neighboring regions are positioned close to a corner of the pRb AB cleft. This corner includes a pair of loops. One of them connects α6 and α7 of A box, while the other connects α15 and α16 of B box.

## Methodologies

In this study, we used computational techniques to understand the molecular determinants of the binding mechanism between LTSV40 and pRb, mediated by the LxCxE motif. Through Molecular Dynamics (MD) simulations and Umbrella Sampling, we evaluated the interaction energies and molecular characteristics of the binding process to pRb of five different LTSV40 models.

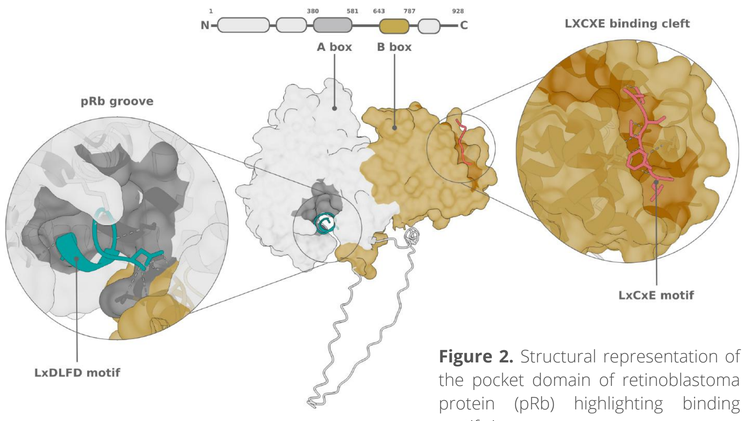


**Figure 2.** Structural representation of the pocket domain of retinoblastoma protein (pRb) highlighting binding motif sites.

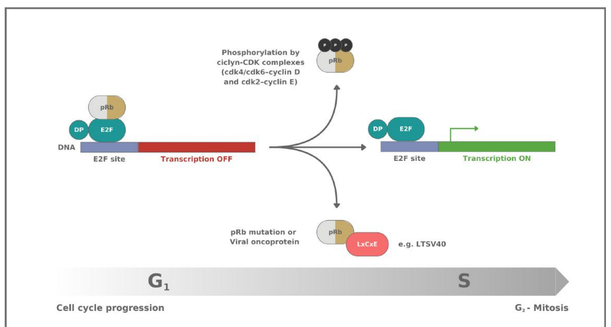


**Figure 3.** Cell cycle regulation in eukaryotes through the interaction of the pRb-E2F complex.

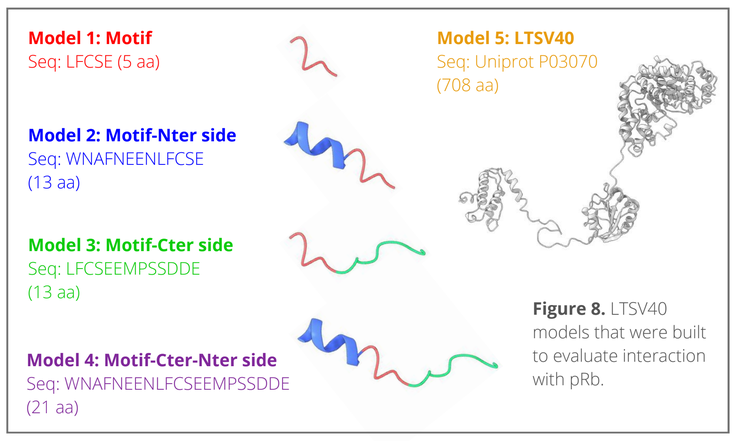


**Figure 8.** LTSV40 models that were built to evaluate interaction with pRb.

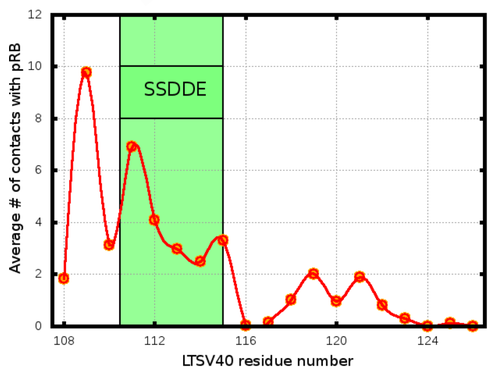


**Figure 7.** Average number of contacts between residues within the sequence Glu108-Pro126 of LTSV40 and any residue of pRb.

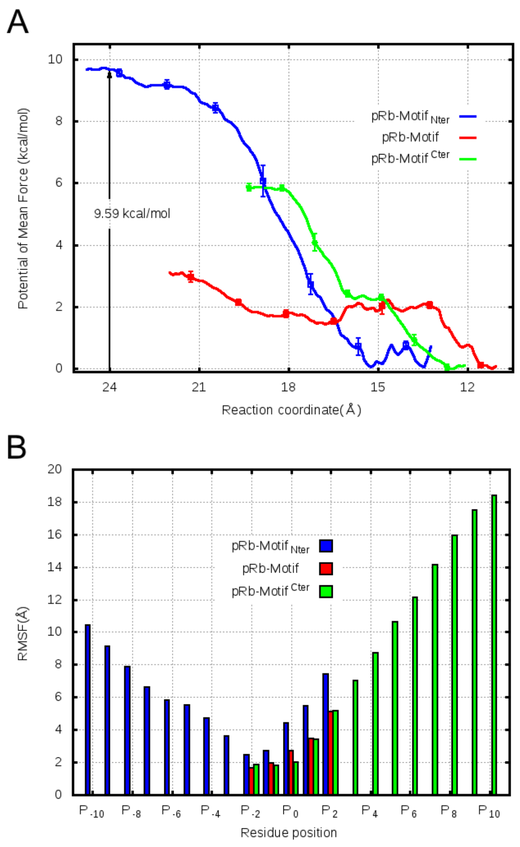


**Figure 9.** Panel A describes the PMF for the binding event of pRb-Motif (red curve), pRb-Motif-Cter side (green curve) and pRb-Motif-Nter side (blue curve). Panel B describes the RMSF values of the Cα atoms that compose the pRb interactor partners for each case.

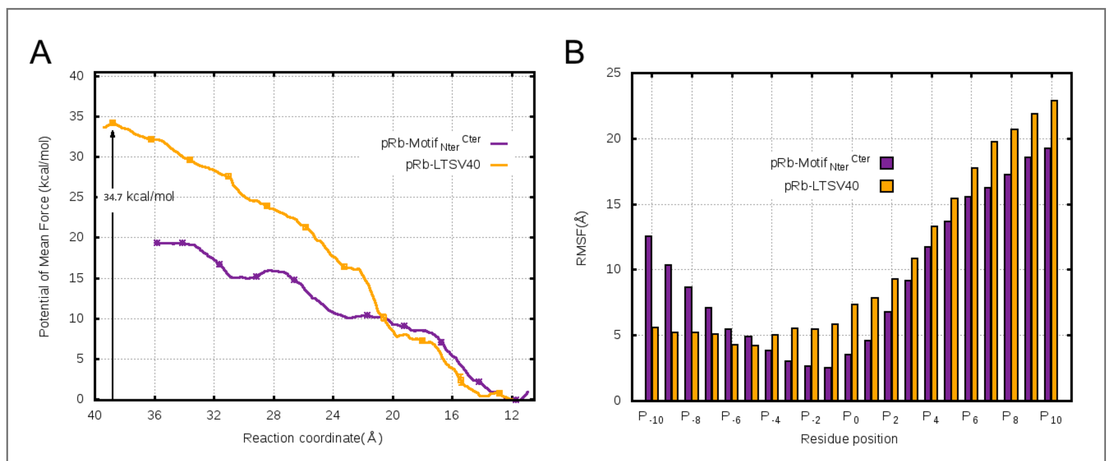


**Figure 10.** Panel A describes the PMF for the binding event of the pRb-Motif-Cter-Nter side (purple) and pRb-LTSV40 (orange). Panel B describes the RMSF values of the Cα atoms that make up the interaction partners of pRb in each case.

## Results

Our results show that the regions immediately flanking the LxCxE motif significantly increase the binding energy to pRb. Additionally, we observed that when a complete LTSV40 model is considered, the interaction with pRb increases compared to models that possess the LxCxE motif and all residues that directly interact with pRb. This suggests that the global structure of LTSV40 modulates the interaction affinity of the segments directly involved in the interaction with pRb. Furthermore, when analyzing the molecular binding mechanism between LTSV40 and pRb, we determined that the N-terminal flanking residues of the LxCxE motif in LTSV40 are the first to recognize the pRb pocket. A detailed analysis of the conformational ensemble of free LTSV40 shows that the side chains of these N-terminal residues to the motif are always available for interaction as they are located in an ordered (folded) region of the system. This recognition strategy involving an ordered region combined with interaction through a contiguous disordered region could be a common feature in proteins containing this type of LxCxE motif, facilitating rapid identification and binding to target proteins.

## Conclusions

These findings provide significant insights into the molecular mechanisms involved in the LTSV40-pRb interaction, highlighting the importance of the flanking sequences in enhancing binding affinity. Additionally, the proposed recognition/binding mechanism results could pave the way for the design of drugs and therapies targeting interactions with pRb in various diseases.

SCAN ME

Molecular dynamics videos of the pRb-LTSV40 binding complex and the free proteins.

**Bibliography**

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- Short linear motif core and flanking regions modulate retinoblastoma protein binding affinity and specificity (Nicolás Palopoli et al.,2018).
- Role of pRb in cell cycle arrest, providing insights into its mechanism of action and interactions with other motifs or proteins (Harrington et al., 1998).
- Review of pRb as a tumor suppressor, highlighting its stabilizing role in the cell cycle and interactions, including with the LxCxE motif (Manning & Dyson, 2011).
- Molecular mechanisms underlying RB protein function, including details on its interaction with specific motifs (Dick & Rubin, 2013).
- Structure of pRb bound to E2F, offering a molecular basis for its regulation and how specific interactions, such as with the LxCxE motif, influence pRb function (Xiao et al., 2003).